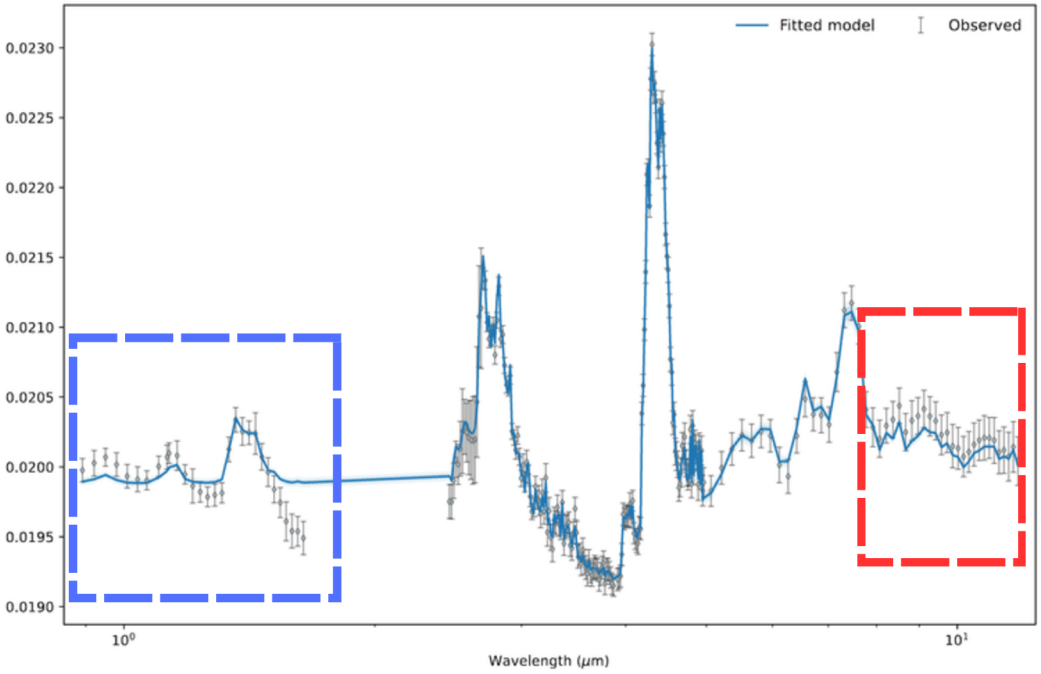
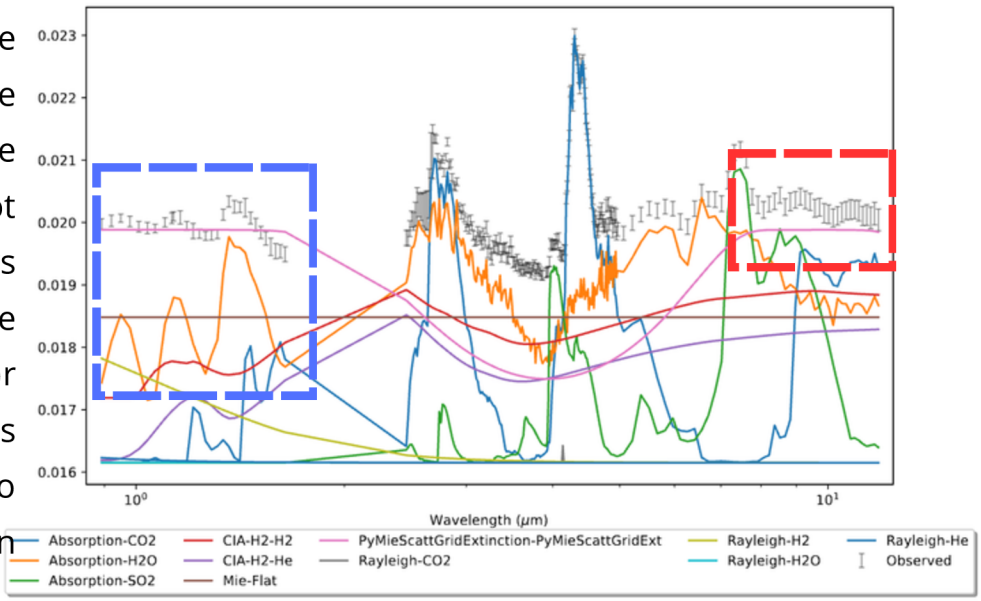
17 - Second iteration

Ahah, did you thought it would be that easy? Getting everything correct on the first try? Nope. I got a few more tricks to show you. Well the fit is already okay but we can improve it. It should look like this from the *output_spectrum.pdf* file created by the **taurex-plot** command:



A good reflex to have is to search which contribution is dominant in the region where the fit isn't good. Luckily for us this command should also have created you a file named: *output_spectrum_contrib_sol0.pdf*. This plot shows the individual contributions:



In the 1 - 2 microns region the retrieval doesn't reproduce the data (blue rectangle). In the same region the contribution plot shows that the main contributors are water and the clouds. These clouds seems also to be the major contribution around 10 microns (red rectangle) where the fit also shows a systematic error. You can also look at the posterior and notice that some parameters seems to reach their priors limits.

Maybe our model for water is not adapted and the clouds are trying to compensate for it, creating these discrepancies. Let's try a *twolayer* profile for water:

```
[[H2O]]
gas_type = twolayer
mix_ratio_surface  = 1e-3
mix_ratio_top = 1e-7
mix_ratio_P = 1e1
mix_ratio_smoothing = 10
```

The fitting parameters for this model are:

- *mol_name_surface* (e.g., *H2O_surface*),
- *mol_name_top*,
- *mol_name_P*.

Do not forget to remove the previous [[H2O]] with a constant mixing ratio as well as the associated fitting parameter or *TauREx* will crash. Also, change the name of the MultiNest folder or it will try to resume the previous run but with different parameters and hence crash. It is good practice to use on par file per retrieval. Also keep in mind to change the name of the output file or the previous one will be deleted by the new.

```
H2O_surface:fit = True
H2O_surface:prior = "LogUniform(bounds=(-12, -1))"
H2O_top:fit = True
H2O_top:prior = "LogUniform(bounds=(-12, -1))"
H2O_P:fit = True
H2O_P:prior = "LogUniform(bounds=(6, -2))"
```

Now launch the second Tatooine retrieval.

...continue to TauREx_demo_3/3.pdf